

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electrical Engineering) SEMESTER - 4 Summer 2025 (Regular)

Course : Bachelor of Technology (Electrical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTBS404 - ANALOG AND DIGITAL ELECTRONICS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 In a transistor amplifier, what is the primary function of the base region?
 - (a) To conduct current between the emitter and collector.
 - (b) To control the flow of current between the emitter and collector.
 - (c) To act as a voltage divider.
 - (d) To provide a high impedance path.
- 2 Which transistor configuration is commonly used for voltage amplification?
 - (a) Common Base
 - (b) Common Collector
 - (c) Common Emitter
 - (d) Common Drain
- 3 Which of the following electrical characteristics is not exhibited by an ideal op-amp?
 - a) Infinite voltage gain
 - b) Infinite bandwidth
 - c) Infinite output resistance
 - d) Infinite slew rate
- 4 An inverting amplifier that amplifies dc input level is called
 - a) DC and AC amplifier
 - b) AC amplifier
 - c) DC amplifier
 - d) None of the mentioned
- 5 What is meant by the parallel load of a shift register?
 - a) All FFs are preset with data
 - b) Each FF is loaded with data, one at a time
 - c) Parallel shifting of data
 - d) All FFs are set with data
- 6 What is the decimal equivalent of the binary number 101101?
 - a) 45
 - b) 43
 - c) 47
 - d) 41
- 7 Which logic gate is also known as an inverter?
 - a) OR
 - b) AND
 - c) NOT
 - d) XOR
- 8 The ability of a logic gate to drive other gates is called:
 - a) Fan-in
 - b) Fan-out
 - c) Propagation Delay
 - d) Noise Margin
- 9 Which of the following expressions is in the sum-of-products form?
 - a) $(A + B)(C + D)$
 - b) $(A * B)(C * D)$
 - c) $A * B * (CD)$
 - d) $A * B + C * D$

